Financial SDEs: Exact Benchmarks and Numerical References

Problem 4 submission presentation

Yifan Chen (2362353) Zeya Hu (2362375)
Yanle Li (2363600) Qingwen Ma (2361052)
Chenyi Xu (2359968)

Numerical Partial Differential Equations

Team number: pending confirmation

Our question: how do we distinguish numerical error from sampling noise?

Two models answer two different questions

GBM: an exact benchmark

- A closed-form solution is available.
- Compare price-based EM and Milstein.
- Check strong and weak convergence.

Non-affine stochastic volatility

- No practical exact transition is available.
- Use EM on log price and volatility factor.
- Check a nested numerical reference.

Takeaway

First verify the method against an exact answer. Then test it without one.

A smaller time step solves only one part of the problem

Strong: compare individual paths

$$e_{\text{strong}}(h) = \mathbb{E}|S_T^h - S_T|$$

Reuse the same Brownian path in both values.

Weak: compare an expectation

$$e_{\text{weak}}(h) = |\mathbb{E}\varphi(S_T^h) - \mathbb{E}\varphi(S_T)|$$

State the observable φ before reporting an order.

sample estimate $-$ target $=$ time-step bias $+$ Monte Carlo noise.

Takeaway

Refining h reduces bias. Increasing the path count reduces sampling noise.

GBM provides an exact, coupled test

$$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad S_T = S_0 e^{(\mu-\sigma^2/2)T+\sigma W_T}.$$

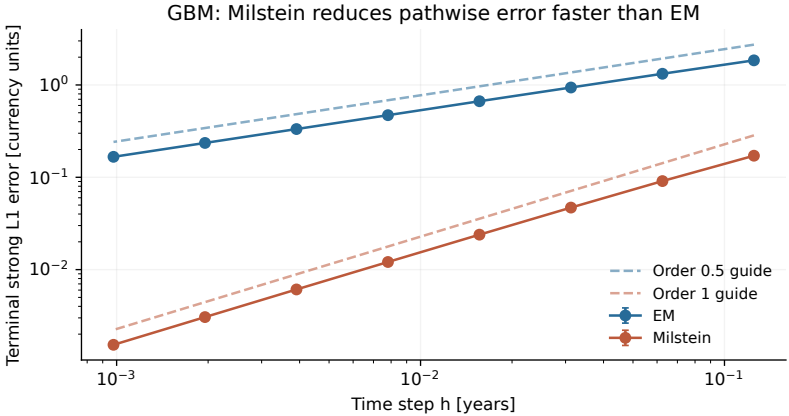
$$\begin{aligned} \text{EM:} \quad & S_{n+1} = S_n(1 + \mu h + \sigma \Delta W_n), \\ \text{Milstein:} \quad & S_{n+1} = S_n\left[1 + \mu h + \sigma \Delta W_n + \tfrac{1}{2}\sigma^2(\Delta W_n^2 - h)\right]. \end{aligned}$$

Self-selected benchmark: $S_0 = 100$, $\mu = .05$, $\sigma = .30$, $T = 1$.
Both methods and the exact endpoint use the same increments; $\Delta W_n \sim N(0, h)$.

Takeaway

The zero-mean Milstein correction improves pathwise accuracy, not every statistic.

GBM strong error supports the expected two orders



Takeaway

Fitted slopes: EM 0.500; Milstein 0.991. Production fit: $N = 64$–1024; 160,000 paths. Independent-seed rate checks are in the Q&A backup.

Why do both GBM weak-mean slopes equal one?

The observable here is the terminal price itself: $\varphi(s) = s$.

The correction has conditional mean zero

$\mathbb{E}[\Delta W_n] = 0$ and $\mathbb{E}[\Delta W_n^2 - h] = 0$, so both updates satisfy

$$\mathbb{E}[S_{n+1} \mid S_n] = (1 + \mu h) S_n.$$

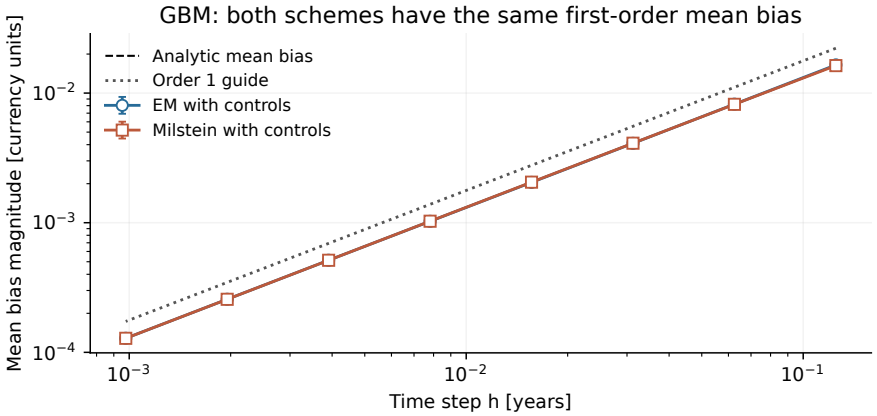
After $N = T/h$ steps, *both* numerical means are $S_0(1 + \mu h)^N$. The exact mean is $m = S_0 e^{\mu T}$, and

$$\mathbb{E}[S_N] - m = -\tfrac{1}{2} m \mu^2 T h + O(h^2).$$

Takeaway

For our nonzero drift, both mean biases are the same and proportional to h. The curves should overlap. This equality is specific to the mean.

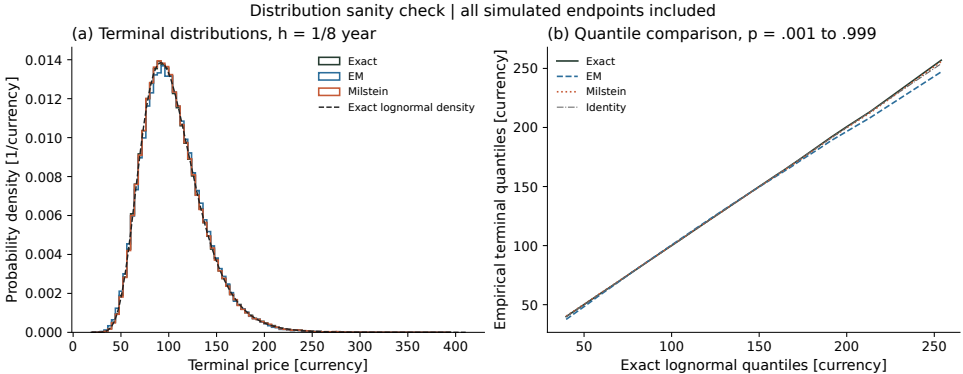
The weak plot confirms identical mean bias



Takeaway

MC slopes 0.999 and 1.000 agree with the analytic order-one bias. Controls are fitted on a separate pilot; plotted estimates are not replaced by the exact bias.

Distribution and positivity provide additional checks

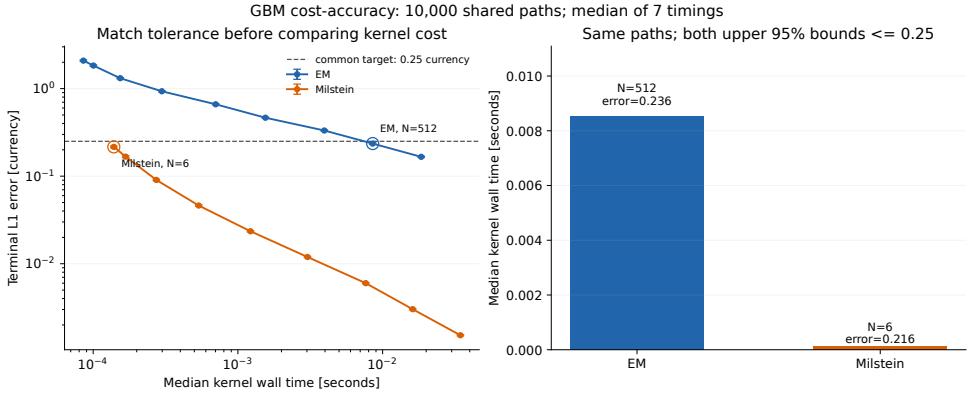


Mean and variance are also checked against the exact lognormal formulas. No negative terminal prices occurred on the reported grids; no clipping was used.

Takeaway

Agreement with the distribution matters. Zero observed negatives does not prove that EM always stays positive.

Compare computational cost at the same accuracy target



Takeaway

Both methods must meet the same error tolerance (0.25 currency units). Cost is kernel-only wall time; actual achieved errors need not be identical.

A non-affine model requires numerical integration

$$\mathrm{d}X_t = \left(\mu - \tfrac{1}{2} g(Y_t)^2\right)\,\mathrm{d}t + g(Y_t)\,\mathrm{d}W_t^{(1)}, \qquad S_t = e^{X_t},$$

$$\mathrm{d}Y_t = \kappa(\theta - Y_t)\,\mathrm{d}t + \xi\sqrt{1 + Y_t^2}\,\mathrm{d}W_t^{(2)}, \qquad g(y) = .1 + \frac{.4}{1 + e^{-y}}.$$

- Y changes the volatility of the log price X.
- Both EM updates use the old value Y_n.
- Correlated increments have prescribed correlation $\rho = -.7$.
- Recovering $S = e^X$ preserves positivity in exact arithmetic; finite, positive states are also checked in the implementation.

Takeaway

We use the problem-pack benchmark. The scalar Milstein order-one claim is not transferred to this two-noise model.

Nested grids reuse the same noise, not the same states



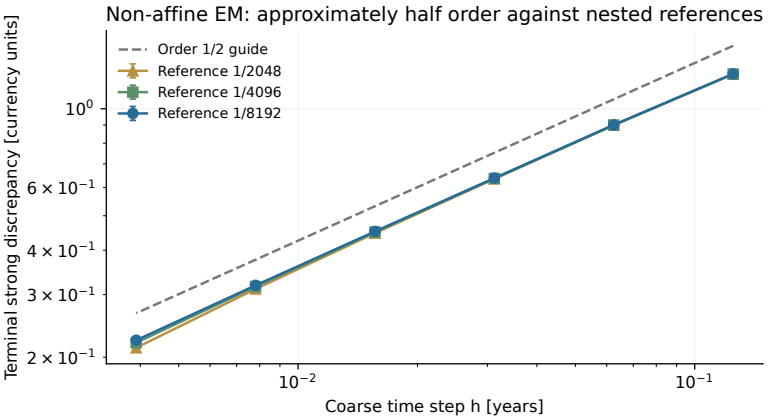
Add consecutive fine increments; rerun EM on each grid.

- Coarse grids: $N = 8$–256; references: 2048, 4096, 8192.
- 100,000 paired paths. The main reference uses 8192 steps.
- Strong discrepancy: $\mathbb{E}|S_T^N - S_T^{8192}|$.

Takeaway

Common noise isolates grid differences. The fine reference is still numerical, not exact.

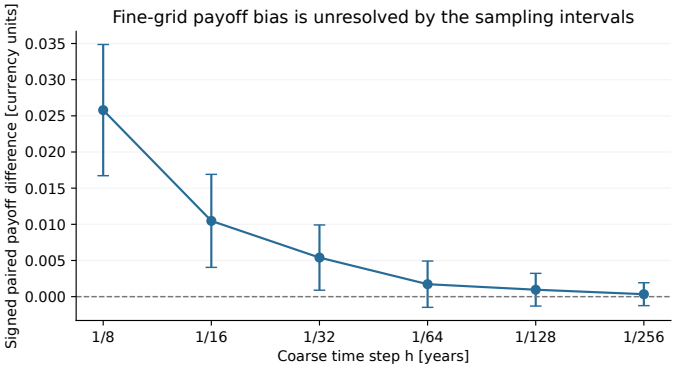
Non-affine strong evidence is stable under refinement



Takeaway

Production slopes 0.496, 0.493, 0.491 use $N = 8, 16, 32, 64$. Two reference refinements support half-order behaviour; they do not bound the unknown exact error.

The nonlinear payoff reveals a weak-resolution limit



Paired payoff: $\varphi(s) = (s - 100)^+$; undiscounted, not a risk-neutral option price.

Takeaway

Fine-grid confidence intervals include zero. We do not claim a reliable weak order. The log-EM mean is exact in expectation, so it cannot test weak convergence here.

Diagnostics and reproducibility support the plots

Check	Evidence
Brownian inputs	Increment variances match h; correlation $-.699675$ matches target $-.7$.
Model states	Finite-state and positivity checks pass on reported grids.
Structural oracle	Non-affine theoretical mean is inside the sampled mean interval.
Reference sensitivity	Three numerical references; two successive refinement tests.
Reproduction	Fixed seeds, recorded environments, one run script, saved tables and checks.

Takeaway

A clear convergence line is useful only when the inputs, reference, and uncertainty are checked.

Main conclusions: accuracy depends on what we measure

1. In GBM, Milstein improves strong order from one-half to one.
2. The two methods have identical first-order weak bias for the mean.
3. Nested-grid log-EM gives stable half-order strong evidence for the non-affine model.
4. The nonlinear weak-payoff rate remains unresolved at the current sampling budget.

Takeaway

Use exact benchmarks where possible. Otherwise refine the reference, couple the paths, and report the limits of the evidence.

Thank you. Questions?

Q&A backup: sources and reproducibility

- Course Problem Pack, Problem 4: Financial SDEs, pp. 6–7.
- D. J. Higham, *An Algorithmic Introduction to Numerical Simulation of Stochastic Differential Equations*, SIAM Review 43(3), 2001, pp. 525–546.
- P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations*, Springer, 1992.

Submission code: `code/run_all.py`, `code/verify.py`, and the supplied README.
This slide is for the three-minute Q&A and is not part of the timed main presentation.

Takeaway

Every numerical conclusion should be traceable to a reproducible experiment.

Q&A backup: why the non-affine mean is not a weak test

Conditional on the old state, $g_n = g(Y_n)$ is known. Log-EM gives

$$S_{n+1} = S_n \exp\left[(\mu - \tfrac{1}{2}g_n^2)h + g_n \Delta W_{1,n}\right].$$

For a Gaussian increment, $\mathbb{E}[e^{g_n \Delta W_{1,n}} \mid \mathcal{F}_n] = e^{g_n^2 h/2}$, so

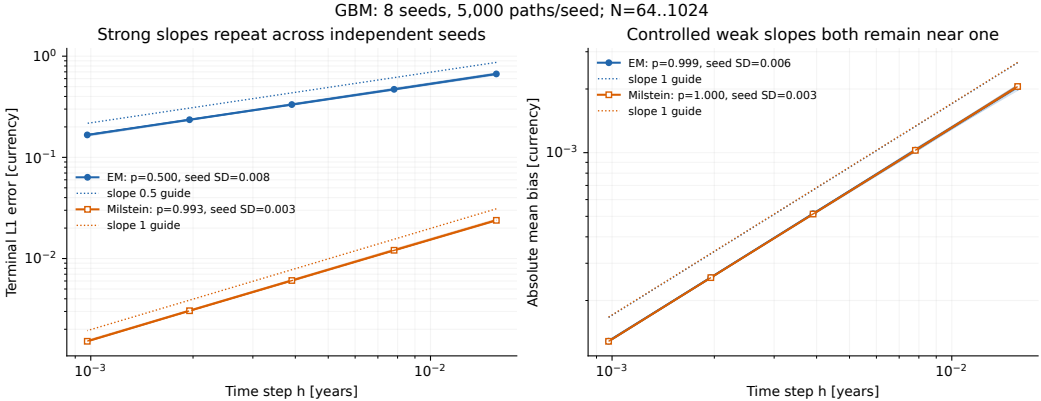
$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n e^{\mu h}, \qquad \mathbb{E}[S_N] = S_0 e^{\mu T}.$$

Correlation between the two drivers does not change this conditional calculation.

Takeaway

An exact mean does not imply an exact terminal distribution. A nonlinear payoff probes a different statistic.

Q&A backup: GBM slopes repeated with independent seeds

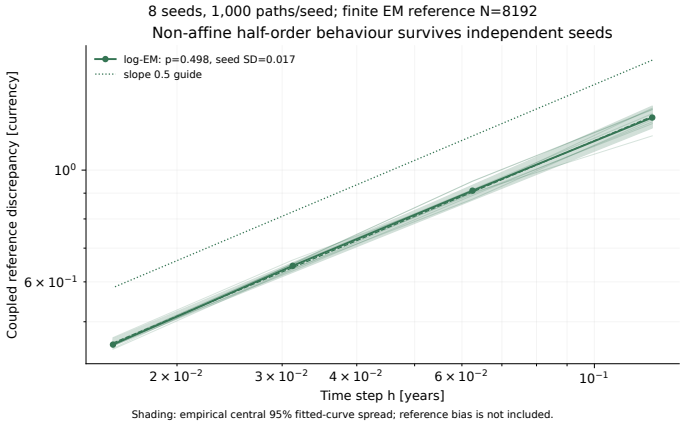


Shading: empirical central 95% spread of independent-run fitted curves, not a guaranteed confidence band.

Takeaway

Eight independent seeds, 5,000 production paths per seed and an independent pilot. Shading shows empirical cross-seed curve spread, not a guaranteed population confidence band.

Q&A backup: the non-affine slope is also repeatable



Takeaway

Eight independent seeds, 1,000 paths each, and the same 8192-step reference protocol. The fitted slope averages 0.498 (cross-seed SD 0.017); this is a smaller-budget repeat check.